

19. Design a C program to implement process synchronization using mutex locks.

AIM:

To design a C program to implement process synchronization using mutex locks.

PROGRAM:

```
#include <stdio.h>
#include <pthread.h>
// Shared variable
int counter = 0;
// Mutex
pthread_mutex_t mutex;
// Function executed by threads
void *threadFunction(void *arg)
{
    int i;
    for (i = 0; i < 1000000; i++)
    {
        pthread_mutex_lock(&mutex);
        counter++;
        pthread_mutex_unlock(&mutex);
    }
    return NULL;
}
int main()
{
    pthread_t thread1, thread2;
    // Initialize mutex
    pthread_mutex_init(&mutex, NULL);
```

```
// Create threads
pthread_create(&thread1, NULL, threadFunction, NULL);
pthread_create(&thread2, NULL, threadFunction, NULL);
// Wait for threads to finish
pthread_join(thread1, NULL);
pthread_join(thread2, NULL);
// Destroy mutex
pthread_mutex_destroy(&mutex);
// Print final counter value
printf("Final counter value: %d\n", counter);
return 0;
}
```

OUTPUT:

Output

Status : Successfully executed

Time:	Memory:
0.0300 secs	1.788 Mb

Your Output

Final counter value: 2000000